NLP Project Guidelines

I. Project Guidelines:

1- Only one member should fill in the form, submit the code, and paper.

2- If you need to update your submission, do not resubmit it. Email the TA with the update and the email of students who filled out the original form.

3- A plagiarism checker will be applied to the code and paper.

4- This project should be either your graduation project or a project from another course.

5- The number of team members allowed is from 3 to 5.

6- You must follow the research paper **template here.**

7- **If you cannot find a team, fill out the form (last question).**

II. Project Time Plan

Task	Deliverables	Deadline
Specify project teams (5 members max) and **choose** project idea.	**Form** Entry that includes (Project Title, Team members).	16/04/26 at 11:59 PM
Phase I	Short **presentation** containing **(5 minutes SHARP):** • Dataset exploration. • Data preprocessing. • Visualization. • Literature review. • Pipeline. • Proposed methodology.	26/4 - 30/4
Phase II (final)	1- **Research paper** that includes Abstract, Introduction, related work, proposed solution, results, discussion and Conclusion.	17/5 - 21/5

	2- **Code** submission. 3- Project **presentation** **(7 minutes SHARP).**	

III. Grading Rubric:

Item	Points
Phase 1	**6**
Data Exploration + Data Preprocessing + Imbalance handling + Visualization	2
Literature Review	2
Pipeline and Proposed Methodology	2
Phase 2	**19**
Different Models Implementation	5
Evaluation Metrics	5
Error Analysis	5
Research Paper + Presentation	5
Individual Oral	5
Total	**25**
Bonus	1 to 5
Total with credit up to	**25 to 30**

IV. Suggested ideas:

For any idea, if a specific dataset or model link is provided, then you must use it.

Advanced Scientific Translation for Arabic Abstracts:

- Develop a translation system for scientific abstracts from English to Arabic using the ASCAT dataset, focusing on preserving meaning, structure, and scientific terminology.
- Apply preprocessing techniques to clean and prepare long scientific texts, handling domain-specific vocabulary and complex sentence structures.
- Build a baseline translation approach and compare it with advanced transformer-based models to improve translation quality.
- Evaluate the translations using standard metrics such as BLEU and analyze how well the system preserves scientific meaning and terminology.
- Investigate common challenges such as ambiguous terms, long-context dependencies, and maintaining consistency in scientific writing style.
- Conduct error analysis to identify issues such as incorrect terminology, loss of meaning, and structural differences between English and Arabic scientific texts.
- **Dataset Link:** https://huggingface.co/datasets/NAMAA-Space/ASCAT-Arabic-Scientific-Translation

Arabic multi-domain aspect-based sentiment analysis system using the A-MASA dataset:

- Apply text preprocessing techniques (tokenization, normalization, regex cleaning) tailored for Arabic (MSA and dialects).
- Followed by extracting linguistic features such as n-grams and POS tags, and optionally leveraging sequence labeling schemes (e.g., BIO) and syntactic information from dependency parsing for aspect extraction.
- Build a baseline model using logistic regression with vector semantics (e.g., TF-IDF or word embeddings) for both aspect extraction and sentiment classification, and compare it with different models: classical classifiers, neural language models (e.g., RNN/LSTM), and fine-tuned transformer models to perform aspect-based sentiment analysis across multiple domains.
- The project should include analyzing how different representations (traditional vs embeddings vs transformers) and domain diversity (seen vs unseen domains) impact performance, while evaluating results using standard metrics such as Precision, Recall, and F1-score.
- **Dataset link:** https://github.com/YomnaEid/A-MASA-Arabic-Multidomain-Aspect-based-Sentiment-Analysis-Datasets

Audio-Based Domain Classification using the NileTTS Dataset:

- Process raw audio signals from the NileTTS dataset by applying standard audio preprocessing techniques such as resampling, normalization, silence trimming, and noise reduction to ensure consistent input quality.
- Extract meaningful acoustic features from speech, including Mel-Frequency Cepstral Coefficients (MFCCs), spectrogram representations, and optionally deep speech embeddings using pretrained models such as wav2vec 2.0 or Whisper.
- Build a baseline classification model using traditional machine learning approaches (e.g., Support Vector Machines or Random Forest) trained on handcrafted audio features.
- Develop advanced models using deep learning architectures such as Convolutional Neural Networks (CNNs) or transformer-based encoders to capture temporal and contextual patterns in speech signals.
- Train the system to classify each audio sample into its corresponding domain (e.g., medical, sales, or general conversation), and evaluate model performance using standard metrics such as accuracy, precision, recall, and F1-score.
- Conduct a comparative analysis between traditional feature-based models and deep learning approaches to understand the impact of representation learning on classification performance.
- **Dataset Link:** https://huggingface.co/datasets/KickItLikeShika/NileTTS-dataset

Multimodal Domain Classification using Audio and Text using NileTTS Dataset:

- Develop a multimodal classification system that leverages both audio signals and their corresponding text transcripts to identify the domain of Egyptian Arabic speech (e.g., medical, sales, general conversation).
- Apply audio preprocessing and extract acoustic features such as MFCCs, spectrograms, or deep embeddings (e.g., wav2vec/Whisper), alongside text preprocessing and extraction of linguistic features using pretrained Arabic language models.
- Design and implement models for three settings: audio-only, text-only, and multimodal fusion,

where both modalities are combined using feature-level or model-level fusion techniques.
- Train and evaluate the models using standard classification metrics such as accuracy, precision, recall, and F1-score, ensuring consistent experimental settings across all approaches.
- Conduct a comparative analysis to assess the contribution of each modality, and investigate how multimodal learning enhances performance, particularly in challenging or ambiguous conversational scenarios.
- **Dataset Link:** https://huggingface.co/datasets/KickItLikeShika/NileTTS-dataset

Audio-Based Medical Named Entity Recognition using NileTTS Dataset:

- Develop an audio-based medical information extraction system that identifies and classifies important medical entities directly from spoken Egyptian Arabic, such as symptoms, diseases, medications, body parts, and clinical conditions.
- Process the speech recordings using audio preprocessing techniques such as resampling, normalization, silence removal, and noise reduction, then extract relevant acoustic features or deep speech embeddings using models such as wav2vec 2.0 or Whisper.
- Convert speech into textual or intermediate representations and apply named entity recognition methods to detect medical terms mentioned in the audio conversations.
- Build and compare different approaches, including a baseline pipeline based on automatic speech recognition followed by text-based NER, and more advanced end-to-end or hybrid systems that combine speech and language understanding.
- Evaluate the system using standard information extraction metrics such as precision, recall, and F1-score, with particular attention to how accurately medical entities are recognized in dialectal and conversational speech.
- Analyze common error cases, such as misrecognized medical terminology, dialectal variations, and ambiguity in spoken descriptions, to better understand the challenges of medical entity extraction from speech.
- **Dataset Link:** https://huggingface.co/datasets/KickItLikeShika/NileTTS-dataset

Speech-to-Text Question Answering End-to-End System using NileTTS Dataset:

- Speech → Text → QA Pipeline (End-to-End system).
- Build an end-to-end system that takes Egyptian Arabic speech as input, converts it into text using Automatic Speech Recognition (ASR), and then applies a Question Answering (QA) model to answer questions based on the transcribed content.
- Use the NileTTS audio data to generate transcripts, then construct question-answer pairs from the transcript content, such as identifying the speaker's intent, extracting important details, or answering domain-specific questions.
- Train and compare QA models on the generated transcripts, and evaluate the full pipeline using ASR metrics such as Word Error Rate (WER) and QA metrics such as accuracy, Exact Match (EM), and F1-score.
- Analyze how transcription errors affect downstream question answering performance, and compare results using automatic transcripts versus reference transcripts to understand the impact of speech recognition quality on the overall system.
- **Dataset Link:** https://huggingface.co/datasets/KickItLikeShika/NileTTS-dataset

Audio-Based Intent Classification using NAMAA-Egyptian-TTS Dataset:

- Develop an audio-based intent classification system for Egyptian Arabic by first constructing a labeled speech dataset using the NAMAA-Egyptian-TTS model, where students define a set of intents such as inquiry, request, complaint, greeting, and confirmation, then create corresponding dialectal text samples and convert them into speech.

- Apply standard audio preprocessing techniques such as resampling, normalization, and silence trimming, then extract acoustic features including MFCCs, spectrograms, or deep speech embeddings using pretrained models such as wav2vec 2.0 or Whisper.
- Build a baseline intent classification model using traditional machine learning approaches such as Logistic Regression or Support Vector Machines, and compare it with advanced deep learning models such as CNNs or transformer-based architectures for speech classification.
- Train the system to classify each audio sample into its corresponding intent category, and evaluate performance using standard metrics such as accuracy, precision, recall, and F1-score, supported by confusion matrices and class-level analysis.
- Conduct error analysis to investigate common challenges such as confusion between similar intents, the effect of synthetic speech quality, dialectal variation, and pronunciation inconsistencies, in order to better understand the strengths and limitations of using generated Egyptian Arabic speech for downstream NLP tasks.
  - **Dataset Link:** https://huggingface.co/NAMAA-Space/NAMAA-Egyptian-TTS

> Note: for building the intent labeled data, you can follow on the following approaches:

- **Rule-Based Labeling:**

You can define a small set of intents, e.g.:

- Inquiry
- Complaint
- Request
- Greeting
- Confirmation

Then Write templates like:

  - "عايز أعرف..." → Inquiry
  - "المشكلة إن..." → Complaint
  - "ممكن..." → Request

Then Match keywords or patterns in text

- **LLM-Based Labeling:**

Prompt example: Classify the intent of the following Egyptian Arabic sentence into: {Inquiry, Complaint, Request, Greeting, Confirmation}

Sentence: "عايز احجز معاد عند الدكتور"
Answer: Request

Dialogue State Tracking using Saudi Najdi Dialect Conversations Dataset:

- Develop a Dialogue State Tracking system that extracts and updates structured information from multi-turn Saudi dialect conversations, such as user intent, entities, and key attributes across dialogue turns.
- Apply preprocessing techniques for Saudi dialect Arabic, including normalization of informal spelling and handling dialect-specific expressions.
- Define a dialogue state schema in a slot-filling format and build models that track and update this state as the conversation progresses.
- Compare baseline approaches with advanced context-aware models, and evaluate performance using metrics such as slot accuracy, joint goal accuracy, and F1-score.
- Analyze challenges such as ambiguity, context dependency, and dialectal variation to understand their effect on dialogue state tracking performance.

- **Example:** "I need to reserve an appointment with the doctor tomorrow."
    - Intent: booking
    - Entity: hospital
    - Time: tomorrow
- **Dataset Link:** https://huggingface.co/datasets/HeshamHaroon/saudi-dialect-conversations

Response Generation using Saudi Dialect Conversations Dataset:

- Develop a conversational AI system that generates appropriate responses for multi-turn Saudi dialect dialogues by learning from the conversational patterns in the dataset.
- Apply text preprocessing tailored to Saudi dialect Arabic, including normalization of informal spelling, handling dialect-specific expressions, and cleaning conversational noise.
- Construct input–output pairs where the model takes the dialogue history as input and generates the next response, enabling context-aware response generation.
- Build a baseline model using simple sequence-to-sequence approaches, and compare it with advanced models such as transformer-based architectures or fine-tuned large language models.
- Train the system to produce coherent, contextually relevant, and fluent responses that align with the conversation flow and user intent.
- Evaluate the generated responses using both automatic metrics (e.g., BLEU, ROUGE) and qualitative human evaluation focusing on relevance, coherence, and naturalness.
- Conduct error analysis to identify issues such as generic responses, loss of context, and dialectal inconsistencies, and analyze how these challenges affect conversational quality.
- **Dataset Link:** https://huggingface.co/datasets/HeshamHaroon/saudi-dialect-conversations

Dialogue Topic Classification for Saudi Dialect Conversations

- Develop a dialogue topic classification system that identifies the main topic or multiple topics discussed in Saudi dialect conversations, using the conversational content across single turns or full multi-turn dialogues.
- Apply text preprocessing techniques tailored for Saudi dialect Arabic, including normalization of informal spelling, handling dialect-specific vocabulary, and cleaning conversational noise to improve text consistency and model performance.
- Formulate the task as a multi-label classification problem, where each conversation may belong to one or more topic categories, and prepare the dataset accordingly at either the utterance level or the full-conversation level.
- Build a baseline model using traditional text classification methods such as TF-IDF with Logistic Regression or Support Vector Machines and compare it with advanced transformer-based Arabic language models that better capture contextual and semantic information.
- Train and evaluate the models using appropriate multi-label evaluation metrics such as precision, recall, F1-score, and subset accuracy, while analyzing the ability of the models to distinguish overlapping or closely related topics.
- Investigate topic variation across dialogue turns, including whether the conversation remains focused on one topic or shifts across multiple topics as the dialogue progresses.
- Conduct error analysis to identify challenges such as topic overlap, ambiguous expressions, short utterances, and dialectal variation, and assess how these factors affect topic classification accuracy in real conversational settings.
- **Dataset Link:** https://huggingface.co/datasets/HeshamHaroon/saudi-dialect-conversations

Dialogue Difficulty Modeling using Saudi Dialect Conversations Dataset:

- Develop a system that predicts the difficulty level of Saudi dialect conversations, such as simple, medium, or complex, based on linguistic and contextual properties of multi-turn dialogues.

- Apply preprocessing techniques for Saudi dialect Arabic, including normalization of informal spelling and handling dialect-specific expressions.
- Model difficulty using features such as dialogue length, topic shifts, ambiguity, and contextual dependency, and compare traditional feature-based methods with advanced transformer-based models.
- Evaluate the system using metrics such as accuracy and F1-score, and analyze how conversational complexity affects model performance.
- Conduct error analysis to study difficult cases such as implicit meaning, overlapping intents, and rapid topic changes.
- **Dataset Link:** https://huggingface.co/datasets/HeshamHaroon/saudi-dialect-conversations

Long-Document Classification using AraModernBERT:

- Develop a document classification system that leverages AraModernBERT's long-context capability to process Arabic texts exceeding traditional BERT limits.
- Compare performance between:
  - truncated input (512 tokens)
  - full long-context input (up to 8k tokens)
- Train models for tasks such as topic classification or document categorization and evaluate using accuracy and F1-score.
- Analyze how long-context modeling improves understanding of full documents and reduces information loss.
- Note: you must compare this model's performance to other models based on the literature review you made.
- **Model Link:** https://huggingface.co/NAMAA-Space/AraModernBert-Base-V1.0

Arabic Document Retrieval System using AraModernBERT

- Develop a semantic document retrieval system that uses AraModernBERT to encode Arabic queries and documents into dense vector representations for effective similarity-based search.
- Apply preprocessing techniques tailored for Arabic text, including normalization, tokenization, and handling of dialectal variations to improve representation quality.
- Build a retrieval pipeline where both queries and documents are embedded using AraModernBERT, and relevant documents are retrieved based on similarity measures such as cosine similarity.
- Implement a baseline retrieval approach using traditional methods such as TF-IDF or BM25, and compare it with the embedding-based neural retrieval system.
- Evaluate the system using standard information retrieval metrics such as Precision@k, Recall@k, and Mean Reciprocal Rank (MRR), focusing on ranking quality and relevance of retrieved results.
- Analyze system performance across different query types, including short queries, long queries, and ambiguous queries, to understand the strengths and limitations of semantic retrieval in Arabic.
- Conduct error analysis to investigate cases where the system retrieves irrelevant or partially relevant documents, and assess the impact of linguistic variation, synonymy, and contextual ambiguity on retrieval performance.
- Note: you must compare this model's performance to other models based on the literature review you made.
- **Model Link:** https://huggingface.co/NAMAA-Space/AraModernBert-Base-V1.0

Parenting Advice Question Answering System using Book- or Blog-Based Retrieval

- Develop a question answering system that provides parenting advice based on a collection of parenting books or parenting blogs, ensuring that responses are grounded in trusted source material rather than generated freely.
- Design the system to support multiple parenting approaches, such as positive parenting, prophetic parenting, gentle parenting, or other educational and behavioral frameworks, allowing the user to select the preferred parenting methodology before asking a question.

- Organize the source material according to parenting approach, so that the system retrieves answers only from the selected methodology and generates advice that is consistent with its principles and guidance style.
- Apply text preprocessing techniques such as cleaning, normalization, and segmentation of books or blog articles into meaningful passages for efficient retrieval and answer generation.
- Represent text using traditional methods such as TF-IDF as a baseline and compare them with embedding-based approaches using pretrained transformer models to improve semantic retrieval quality.
- Implement a retrieval mechanism that identifies the most relevant passages for a user's parenting question, then generate the response by extracting or summarizing advice from the retrieved content while preserving alignment with the chosen parenting approach.
- Train and evaluate the system by comparing retrieval methods and analyzing how the selected parenting methodology affects the relevance, consistency, and usefulness of the generated answers.
- Conduct error analysis to investigate challenges such as overlapping advice across parenting approaches, ambiguous user questions, conflicting recommendations between methodologies, and retrieval errors from books versus blogs.

Your own idea:

Email your course professor and TA with the project abstract to get approval of your own idea.